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**ON THE OOGAMOUS SEXUAL REPRODUC-
TION IN A CARTERIA**

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ON THE OOGAMOUS SEXUAL REPRODUCTION IN A CARTERIA*

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(Communicated by M. O. P. Iyengar)

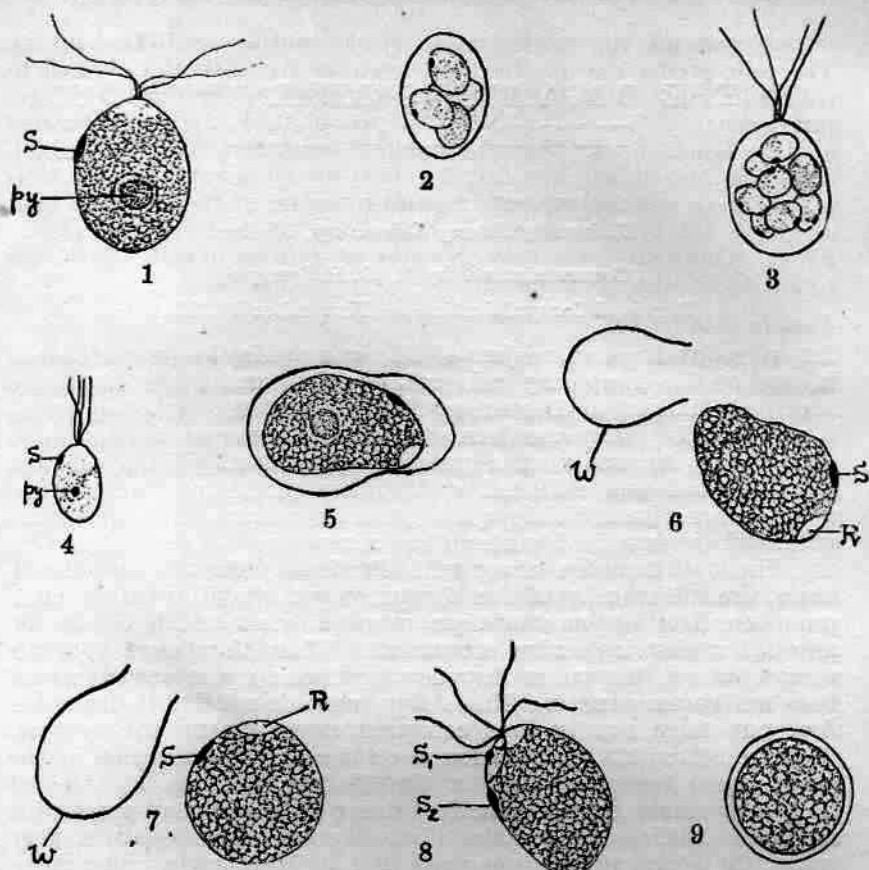
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THE alga forming the subject of this communication was collected in one of the temporary rain-water pools formed on the sandy beach at Madras in December 1940, during the N.E. monsoon. It was found in large numbers on moist sand very near the water edge giving the wet sand a greenish appearance. When the wet greenish sand was collected and shaken with some water, the water became immediately quite green in colour. A drop of this water when examined under the microscope showed numerous individuals of the alga along with *Pandorina*, *Scenedesmus*, *Cælastrum*, etc. The cells of the alga were quite healthy and bright green in colour, but most of them were quiescent and had no cilia, only a few showing active movement. When they were kept for a short time in the water they developed cilia again and began to show active movement. These algæ were found inside the pool also, but their number was comparatively very small. Iyengar (1920, p. 333 ; 1933, p. 324) found on the moist sand fringing the temporary rain-water pools in the Madras beach a number of unicellular and colonial Volvocales, giving the moist sand a greenish appearance. He suggested that these algæ preferred this situation presumably due to the greater aeration and the lower temperature of the wet sand than inside the pools.

VEGETATIVE CELLS

The cells of the alga are ovate to ellipsoid in shape and rounded both at the anterior and the posterior ends (Text-fig. 1). Occasionally the anterior end is somewhat slightly truncate. The cell membrane is thin and smooth and closely enveloping the protoplast. No papilla could be seen. The cilia are as long as, or slightly longer than, the length of the body. The shape of the chromatophore could not be made out clearly owing to the dense nature of the chromatophore and also the accumulation of starch inside, but appeared to be somewhat bell-shaped with a small anterior opening. A large median pyrenoid is embedded at its posterior end. A bright linear eyespot is situated a little from the anterior end (Text-fig. 1). Two contractile vacuoles are seen just below the insertion of the cilia. The nucleus is located just above the median portion inside the hyaline cytoplasm.

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Text-figs. 1-9.—*Carteria Iyengarii* sp. nov. Fig. 1. Vegetative cell. Fig. 2. Four male gametes formed in a cell. Fig. 3. Eight male gametes formed inside a cell which still retains its cilia. Fig. 4. A male gamete. Fig. 5. A female cell showing the gelatinization of the wall at the anterior region and the egg about to escape out. Fig. 6. Egg just escaped out of its mother wall by amoeboid movement. Fig. 7. Egg rounded after escaping from the mother wall. Fig. 8. Fusion of the male gamete with the egg: note the two eyespots of the gametes and the four cilia of the male gamete in the zygote. Fig. 9. Zygote surrounded with a wall. (All figures $\times 950$.) (s, eyespot; s_1 & s_2 , eyespots of the male and the female gamete, respectively; r, receptive spot; w, discarded wall of female gamete; py, pyrenoid.)

REPRODUCTION

Asexual reproduction was not observed in the material. But sexual reproduction takes place through the fusion of a motile quadriciliate antherozoid with a non-motile egg.

Male gametes:

Some of the vegetative cells after coming to rest and losing their cilia form two, four or eight male gametes which finally escape

outside through the gelatinisation of the mother wall (Text-fig. 2). The cells giving rise to the male gametes are somewhat brown in colour in contrast to the bright green colour of the ordinary vegetative cells. These male gametes are pear-shaped or slightly rounded and are much smaller than the smallest vegetative cell (Text-fig. 4). They are four-ciliated and bright golden brown in colour and possess a pale green chloroplast with a small pyrenoid at the posterior end. A bright red eyespot is seen a little from the anterior end (Text-fig. 4). Quite often the male gametes are formed in cells which still retain their cilia (Text-fig. 3).

Female gametes :

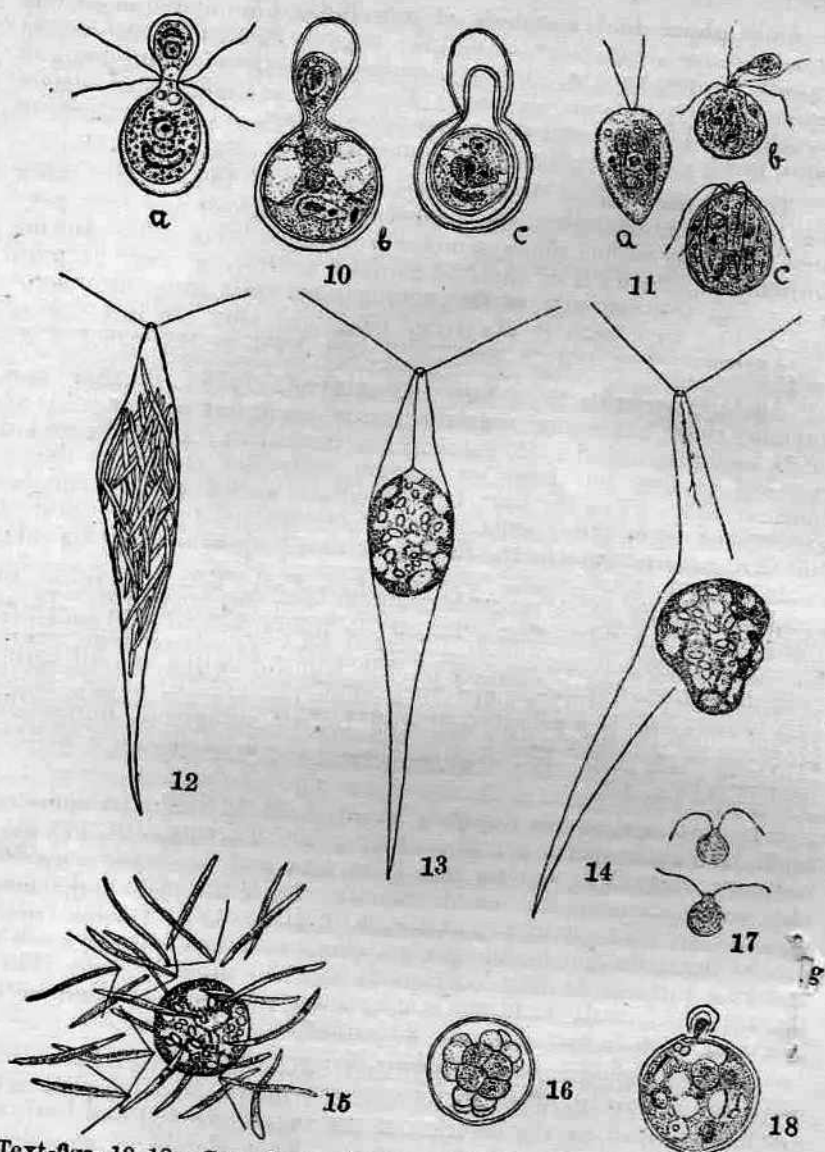
In contrast to the male gamete, the female gamete is formed by the transformation of the entire contents of a single vegetative cell into a large non-ciliated egg (Text-figs. 5-7). The cell giving rise to the egg, after coming to rest, loses its cilia and becomes more bright green in colour. It retains its eyespot, contractile vacuoles and cell-membrane.

Fertilization :

The male gametes, soon after their escape from the mother cell, swim towards the female cells and swarm round them in large numbers. During this swarming, the wall of the female cell at the anterior region gelatinizes (Text-fig. 5) and the entire contents escape out of the wall as a large naked egg by a somewhat amoeboid movement (Text-fig. 6). After escaping outside it again becomes rounded and in that condition shows clearly the eyespot, the two contractile vacuoles and a somewhat hyaline region at the anterior end representing the receptive spot (Text-fig. 7). One of the motile male gametes finally fuses with the naked egg at the region of the receptive spot (Text-fig. 8). The fusion is very rapid, the entire process not exceeding 10-15 seconds. The fertilized egg shows the two eyespots of the male and the female gametes and slowly moves about for some time with the aid of the four cilia of the male gamete (Text-fig. 8). The place where the male gamete fused with the female gamete could be seen as a small brownish area in contrast to the bright green colour of the rest of the zygote. Finally the cilia are lost and the zygote settles down and forms a wall round itself. The fully formed zygote is spherical and somewhat brown in colour and possesses a thick smooth hyaline wall (Text-fig. 9). The further fate of the zygote is not known.

DISCUSSION

So far in all the previously investigated species of *Carteria*, the sexual reproduction is either typically isogamous or only slightly anisogamous. According to Pascher (1927, p. 139) extreme heterogamy is not known in *Carteria*. The present alga is, therefore, very interesting in showing an extreme form of heterogamy, in fact definite oogamy, a type of reproduction so far recorded in only a few members of the unicellular Chlamydomonadaceæ.



Text-figs. 10-18.—Sexual reproduction of the heterogamous and oogamous unicellular Chlamydomonadaceae. Fig. 10 (a-c). *Chlamydomonas Braunii* Goroschankin. Stages in conjugation of gametes. Fig. 11 (a-c). *Phyllariochloris striata* (Korschikoff) Pascher. a, vegetative cell; b, fusion of two gametes; c, zygote. Figs. 12-15. *Chlorogonium oogamum* Pascher. Fig. 12. Formation of male gametes. Fig. 13. Formation of the ovum. Fig. 14. Liberation of the ovum. Fig. 15. Fertilization. Figs. 16-18. *Chlamydomonas coccifera* Goroschankin. Fig. 16. Formation of male gametes. Fig. 17. Male gametes. Fig. 18. Fertilization. (Figs. 10, 11, 16-18, redrawn from Pascher, 1927; Figs. 12-15, redrawn from Pascher, 1931.)

Only about four members of unicellular Chlamydomonadaceae show extreme heterogamy or definite oogamy, viz., *Chlamydomonas Braunii* Goroschankin, *Chlamydomonas coccifera* Goroschankin, *Phyllariochloris striata* (Korsch.) Pascher (= *Phyllomonas striata* Korsch.) and *Chlorogonium oogamum* Pascher, all the remaining forms being either isogamous or anisogamous.

In *Chlamydomonas Braunii* (Goroschankin, 1890) eight biciliate, walled microgametes are formed from each cell and four very large, biciliate, walled macrogametes from some other cells. During conjugation, the walls of the two gametes coalesce at their anterior region and the contents of the microgamete pass into the macrogamete and fuse with it (Text-fig. 10). The cilia are lost during conjugation. The zygote secretes a new wall in the usual way.

In *Phyllariochloris striata* (Korschikoff, 1925; Pascher and Jahoda, 1928) the sexual reproduction is somewhat similar. 16, 32 or 64 biciliate, naked male gametes are formed in a cell. These are rounded at first but later on become somewhat elongated like a spermatozoid. Two or four large biciliate walled female gametes are formed from other cells. The zygote formed by the fusion of the two gametes, retains the four cilia for a long time (Text-fig. 11).

In *Chlorogonium oogamum* (Pascher, 1931) a large number of biciliate male gametes is formed from a cell (Text-fig. 12). These are narrow and very much elongated like typical spermatozooids. The female gamete is formed by the transformation of the entire contents of a vegetative cell into a single non-ciliated ovum, which escapes out of the mother-cell-wall by an amoeboid movement (Text-figs. 13, 14). Fertilization takes place outside the mother wall (Text-fig. 15).

In *Chlamydomonas coccifera* (Goroschankin, 1905) 16 spherical biciliate male gametes are formed in a cell (Text-figs. 16, 17). An ordinary vegetative cell by losing its cilia and becoming considerably enlarged forms the female gamete. Both the male and female gametes are clothed with a cell-wall (Text-fig. 18). During conjugation the male and female gamete come together and the walls of the two gametes become contiguous at their anterior ends (Text-fig. 18). The contents of the male gamete pass into the female cell and fuse with it and a zygote is formed.

Of these four forms, in the first two, viz., in *Chlamydomonas Braunii* and *Phyllariochloris striata*, the micro- and macrogametes are both formed by the division of the vegetative cell and both are ciliated until the time of conjugation. They should be considered therefore as cases of extreme heterogamy, very nearly approaching oogamy. On the other hand, in the two latter cases, viz., *Chlorogonium oogamum* and *Chlamydomonas coccifera*, the female gamete, which is formed by the transformation of the entire contents of a single cell without division, is non-ciliated and constitutes a definite ovum or egg. These two cases should therefore be considered as typically oogamous. Of these two again, in *Chlorogonium oogamum*

the egg escapes out of its mother-cell wall by an amœboid movement and is fertilized outside the parent cell-membrane. In the case of *Chlamydomonas coccifera* the egg does not escape outside and fertilization takes place inside the female cell. The sexual reproduction in this alga should be considered as more advanced than in *Chlorogonium oogamum*, since its egg is completely non-motile and does not show even the slight movement shown by the egg of the latter alga when escaping out of its cell membrane. *Chlamydomonas coccifera* should be considered therefore as showing the highest expression of sexual reproduction among the unicellular Chlamydomonadaceæ.

As regards the present alga, its sexual reproduction is very similar to that of *Chlorogonium oogamum*. Here also the entire contents of a single vegetative cell are transformed without division into a single non-ciliated egg which escapes out of its parent wall by an amœboid movement and is fertilized outside it. However, in the present alga, the male gametes, though small, still retain the shape of the ordinary vegetative cells, in contrast to the typically spermatozoid-like male gametes of *Chlorogonium oogamum*. Therefore, the present alga should be considered as slightly less advanced than *Chlorogonium oogamum* in sexual reproduction.

Regarding the specific position of the present alga, it is quite different from all the other species of *Carteria* recorded before, not only in its morphological features, but also in having oogamous sexual reproduction. It is therefore best to consider it as a new species which may be named as *Carteria Iyengarii* sp. nov.

DESCRIPTION

Carteria Iyengarii sp. nov.

Cells ovate-ellipsoid with a broadly rounded posterior end; anterior end broadly rounded or slightly truncate; membrane thin; papilla absent; cilia as long as the length of the body or slightly longer; chromatophore bell-shaped, almost covering the entire cell, with a single median pyrenoid placed in the posterior region; eyespot linear, placed a little distance from the anterior end; contractile vacuoles two, just below the cilia; sexual reproduction oogamous, by the fusion of a small four ciliated male gamete with a large non-ciliated egg; two, four or eight naked male gametes formed inside a cell; male gametes four ciliated, brown in colour, rounded or ellipsoid in shape with an eyespot and a small chromatophore with a pyrenoid; female gametes formed singly by the escape of the entire protoplast as a naked egg; egg non-ciliate, with an eyespot, two contractile vacuoles and a hyaline receptive spot; zygote rounded with a thick smooth wall. Vegetative cells $20.0-25.0\mu$ long and $15.7-20.0\mu$ broad; male gametes $10.0-11.7\mu$ long and 6.7μ broad; zygote $20.0-21.70\mu$ in diameter.

Habitat.—On moist sand near the water edge in a temporary rain-water pool on the sea-coast at Madras, South India.

SUMMARY

An account is given of the sexual reproduction in *Carteria Iyengarii* sp. nov. from Madras. The sexual reproduction is of a high type in being oogamous and takes place by the fusion of a small biciliate motile antherozoid with a large non-ciliated egg, which escapes out of its parent wall. This appears to be the first record of oogamous reproduction in the genus *Carteria*.

In conclusion, I wish to express my great indebtedness to Prof. M. O. P. Iyengar, M.A., Ph.D. (Lond.), F.L.S., for his kind help and guidance in the preparation of the paper.

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ERRATUM

Volume XXI, Nos. 1 & 2, page 135, line 4, of the paper on "Oogamous Sexual Reproduction in a *Carteria*" by K. R. Ramanathan,

For 'biciliate', please read 'quadriciliate'.